

- Provided to the vendor's customer support for help in commissioning or establishing Node provenance;
- Used programmatically to obtain a Node's [Passcode](#) or other information in order to provide a simplified setup flow. Note that the mechanism(s) by which the [Passcode](#) may be obtained is outside of this specification. If the Rotating Device Identifier is to be used for this purpose, the system implementing this feature SHALL require proof of possession by the user at least once before providing the [Passcode](#). The mechanism for this proof of possession, and validation of it, is outside of this specification.

The Rotating Device Identifier is an optional feature for a Node to implement and an optional feature for a Commissioner to utilize. The algorithm used for generating a Rotating Device Identifier SHALL meet the following security and privacy requirements:

1. It SHALL be irreversible in such a way that:
 - a. It SHALL prevent recovery of a unique identifier for the device by entities that do not already have access to the set of possible unique identifiers.
 - b. Leaking of a common key or equivalent could not be used to recover a unique identifier for all devices sharing the common key.
2. It SHALL protect against long-term tracking by rotating upon each commencement of advertising.
3. It SHALL have a total of at least 64 bits of entropy and SHOULD preferably have more, up to 256 bits.
4. It SHALL NOT contain a fixed identifier such as a serial number.

The [Rotating Device Identifier Algorithm](#) below meets these requirements. A Node that implements the Rotating Device Identifier SHALL use either the Rotating Device Identifier Algorithm or a different algorithm which has been approved and verified by the Connectivity Standards Alliance for this purpose and which meets the same set of security and privacy requirements listed above.

The Rotating Device Identifier Algorithm employs a key derivation algorithm that combines a monotonically increasing lifetime counter with a persistent unique per-device identifier.

This persistent unique identifier SHALL consist of a randomly-generated 128-bit or longer octet string which SHALL be programmed during factory provisioning or delivered to the device by the vendor using secure means after a software update.

This persistent unique identifier SHALL be protected against reading or writing over the air after initial introduction into the device, and stay fixed during the lifetime of the device.

NOTE

This persistent unique identifier SHALL NOT be the same as the UniqueID attribute exposed in the Basic Information cluster.

The lifetime counter SHALL be an integer at least 16 bits in size, incremented upon each commencement of advertising, and wrapping when the maximum value is reached.

The Rotating Device Identifier Algorithm is defined as follows: